FitnessMirror Research Plan

Overview:

The 2024 WHO/Lancet Report states that **31.3% (1.8 billion) of adults** do not participate in **physical activity**. How might we provide busy students and professionals with a personalized, low-friction fitness experience that eliminates the barriers of time and procrastination to help them build a lasting workout rhythm?

Problems:

Those who are not able to workout due to tight schedules, mental tiredness, or a lack of built-in accountability, preventing them from maintaining a sustainable workout habit.

Solution:

A **smart mirror** that removes any resistance encountered while using traditional fitness. It does this by being incorporated directly into the user's **environment**. This mirror offers daily support as an accountabilty partner and since it shows the detailed daily behaviours of the user, it can help even those who may have a challenging or stressful lifestyle by providing personalised assistance and proactive feedback based on the users behaviours; unlike a mobile application that would require users to scroll through their daily schedules to find out if an activity has been scheduled or how to do so.

Hypothesis:

We believe by incorporating a seamless exercising buddy to the user’s environment will create a frictionless exercise experience thats convenient and enjoyable for many individuals. Many people choose not to go to their local gym as a result of not having enough time, lack of motivation and/or feeling uncomfortable at the gym. Using the

Mirror eliminates these reasons by offering people the opportunity to receive real-time feedback, have access to guided workouts and receive personalized workout plans to create an experience similar to having a trainer without actually having a personal trainer.

Research Goal:

- **User needs & motivation**
  Understand why people choose (or avoid) home workouts, and what keeps them consistent.
- **Feature priorities**
  Decide which features matter most, like posture correction, personalized plans, or gamification.
- **Interface design**
  Explore how information should be displayed on the mirror (e.g., minimal UI vs. more detailed feedback).
- **Interaction methods**
  Test how users prefer to interact (touch, gesture, voice, or automatic tracking).
- **Physical setup & context**
  Learn where users would place the mirror and how much space they have at home.

Methodologies:

- **Expert Interviews**
  To gain professional insights into fitness coaching and technical feasibility.
- **Concept Testing**

  Present low-fidelity wireframes, to potential users to ensure usability of product.
- **Surveys**

  Distribute online questionnaires to urban professionals who exercise at least twice a week.

- **Usability Studies**

  Observe participants interacting with a prototype while performing a short workout.

Timeline:

- March 30 - May 11

Participants:

Criteria:

- **Age:** 18–35
- **Occupation:** Full-time students or professionals
- **Environment:** dorms, studios, or shared suites
- **Status:** "High-intent" individuals struggling with fitness consistency, time management, or burnout.